

log₁₀cfu/g). *Enterococcus spp* counts were not different between windrowed and decaked litter for the first 3 cycles (average of 5.83 and 5.94 log₁₀cfu/g, respectively), but were approximately 1 log lower for the windrowed compared to the decaked litter in fourth and fifth cycles (5.73 vs 6.98 and 5.80 vs 6.69 log₁₀cfu/g, respectively). These data indicate that litter windrowing does not effectively reduce litter *Enterococcus spp.* counts between flocks compared to decaking.

321 Field trial results of an automated egg picking robot for commercial broiler breeder flocks. Colin Usher*¹, Michael Park¹, Yu Hang He¹; ¹*Georgia Tech Research Institute, Aerospace Transportation Advanced Systems Lab Food Processing Technology Division, Atlanta, Georgia, United States.*

Researchers at Georgia Tech Research Institute previously demonstrated an egg picking robot in a commercial broiler breeder house. To demonstrate the viability of this prototype as a commercial-ready system, they have benchmarked the performance via an extended field test. The goal is to demonstrate the commercial readiness of the technology and quantify robot performance. A prototype robot was operated fully autonomously in a set of field trials in a broiler breeder facility. In the first set, the robot was operated with researchers observing operation, interfering in the case of undesired behavior from the autonomy algorithms. For the second set, the robot was operated for varying lengths of times unattended. During these times, the ability of the robot to operate without intervention was evaluated. The robot consists of a COTS chassis fitted with components including

a CPU, motors and battery management system, a robot arm, lidar, 3D camera, and a localization system. Algorithms and methods were developed to identify floor eggs and enable the robot to approach and pick them. Commercial-ready functionality was developed including automated docking and charging, and a scheduler allowing for timed deployment of the robot. For the first set of field trials, the robot was operated fully autonomously while being observed by the researchers. During this time, egg detection accuracy was manually tabulated by observing the presence of floor eggs and the robots ability to detect them. The average time to pick an egg was also calculated by recording and tabulating the time it took the robot to pick the eggs once detected. For the second set of trials, the robot was left to operate for extended periods of time, utilizing the scheduling functionality. During this time, the robot automatically executed navigation, docking and charging, and egg picking operations. The time between manual interventions was logged to establish a baseline for automated activity capabilities. Preliminary results indicate a detection accuracy of 85%, a pick success rate of 91% and an average time to pick of 1 minute. Successful demonstration of docking/charging was also observed, but results of extended hands-off operation is still pending. In conclusion, the researchers deployed a robot capable of autonomously navigating a broiler breeder flock and removing floor eggs. A high success rate for picking accuracy was observed.

322 Not Presented.